

C. Ultimate Value

time limit per test: 2 seconds

memory limit per test: 256 megabytes

Let's define a function $f(a)$ for an array a of length n as

$$f(a) = \text{cost} + (a_1 - a_2 + a_3 - a_4 \cdots a_n)$$

where cost is zero initially.

Now consider the scenario where Alice and Bob are given an array a of length n . They play a game taking at most 10^{100} turns alternately with Alice going first.

In each turn, they must perform any **one** (only one) of the following operations:

- End the game for both Alice and Bob.
- Choose two indices l, r with $1 \leq l \leq r \leq n$ and swap a_l and a_r ; this adds $(r - l)$ to the cost.

Assume that Alice tries to maximize $f(a)$ and Bob tries to minimize it.

Your task is to determine the final value of $f(a)$ assuming both players play optimally.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \leq n \leq 2 \cdot 10^5$) — the length of the array a .

The second line contains n integers $a_1, a_2, a_3, \dots, a_n$ ($1 \leq a_i \leq 10^9$) — the elements of the array a .

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each testcase, output a single integer — the final value of $f(a)$ assuming both players play optimally.

Example

input	Copy
<pre>5 2 1000 1 5 9 9 9 9 9 4 7 1 8 4 6 1 14 1 14 1 15 9 31 12 14 22 89 6 78 25 91</pre>	
output	Copy
<pre>999 13 12 -7 265</pre>	

Note

For the first testcase, it is optimal for Alice to end the game on her first turn.

So the final value of cost = 0 and $f(a) = 0 + 1000 - 1 = 999$.

Codeforces Round 1049 (Div. 2)

Contest is running

00:04:40

Contestant



→ Submit?

Language:

Python 3.13.2

Almost always, if you send a solution on PyPy, it works much faster

Choose file:

Choose File

No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	270
Problem B	540
Problem C	675
Problem D	945
Problem E1	945
Problem E2	540
Problem E	1485
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 01:55 from the first attempt

For the fourth testcase, it is optimal for Alice to swap a_1 and a_6 , and then it is optimal for Bob to end the game on his first turn.

So the final value of cost = 5 and $f(a) = 5 + 15 - 14 + 1 - 14 + 1 - 1 = -7$.

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